29-2 NAT Troubleshooting – Answer Key

In this lab you will troubleshoot Network Address Translation.

NAT Troubleshooting

Your company has bought the range of public IP addresses 203.0.113.0/28 for use in a new branch office. The office is not in use yet.
The Service Provider default gateway SP1 uses IP address 203.0.113.1.
203.0.113.2 is to be assigned to the Internet facing F0/0 interface on your router R1.

A junior network administrator was tasked with configuring NAT as follows:

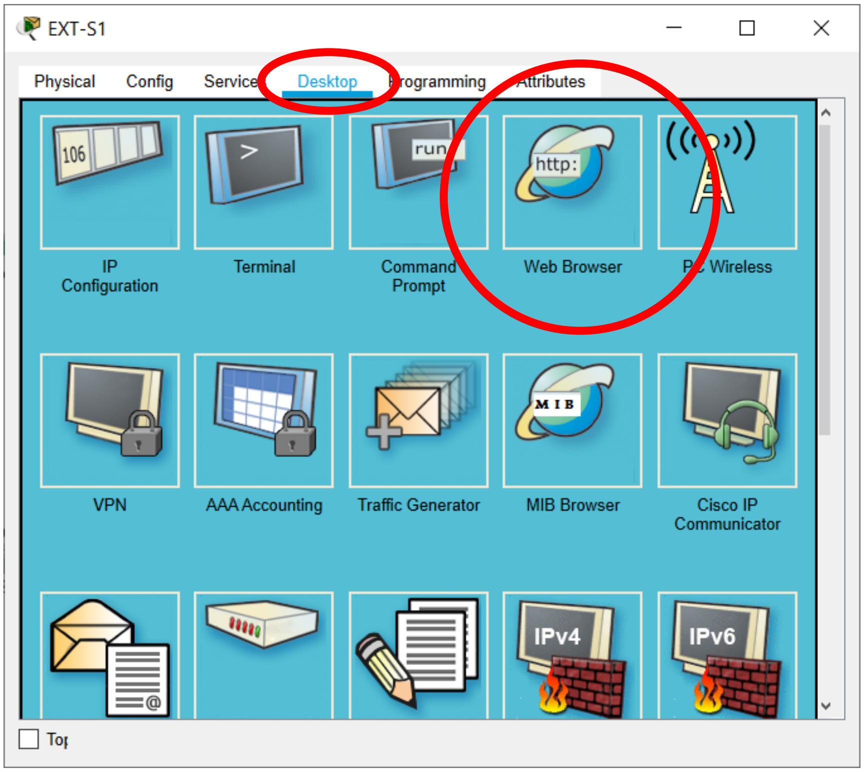
1. A permanent translation should be set up so that external hosts on the Internet can reach the INT-S1 web server using IP address 203.0.113.3. Test this by Telnetting to port 80 on the server from EXT-S1.
2. The 50 internal hosts on the 10.0.2.0/24 internal network should be able to use the remaining public IP addresses to communicate with external hosts. Internal hosts should share public IP addresses where necessary. Test this by pinging EXT-S1 from both PC1 and PC2.

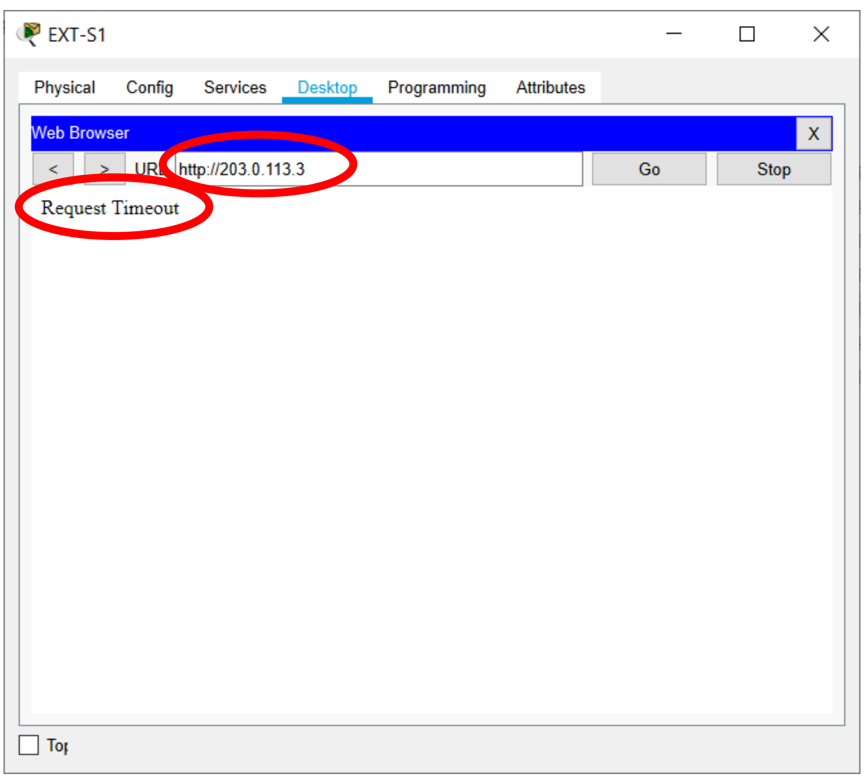
It has been reported that inbound connectivity from the Internet to INT-S1 is not working. INT-S1 and the PCs on the 10.0.2.0/24 network do however have outbound connectivity to the Internet.

Your task is to find the error, explain to the junior network administrator what their mistake was, and then fix the problem.

The task is complete when you can explain why the current configuration is not working, and you have replaced it with a working configuration.

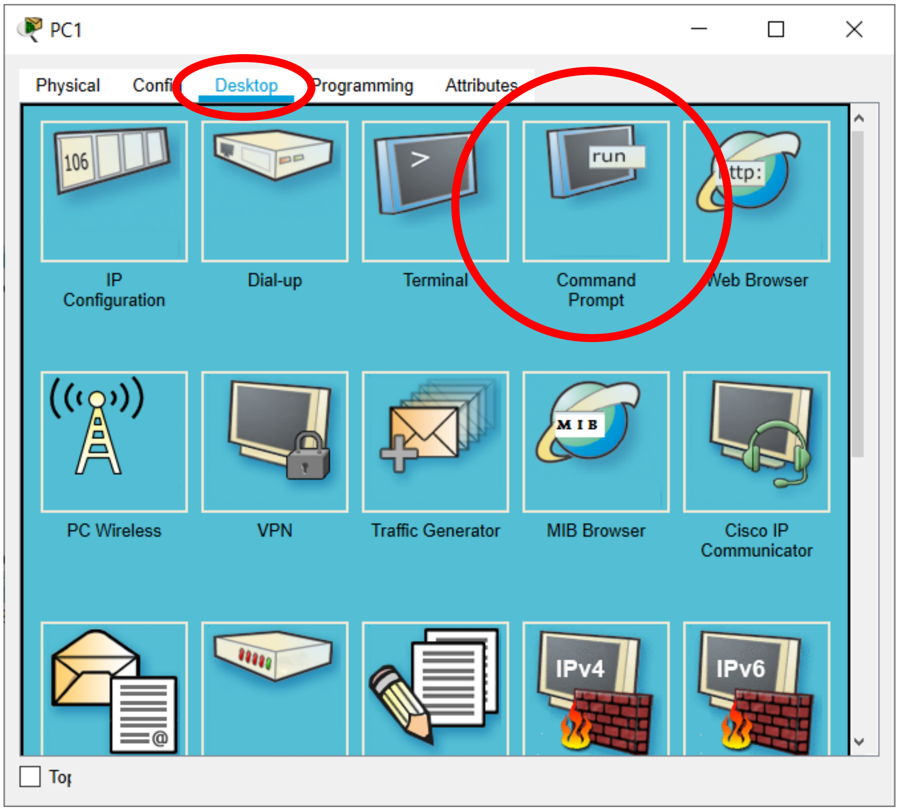
The first thing to do is verify if the current NAT setup is working. Attempt to open an HTTP session from EXT-S1 to INT-S1.

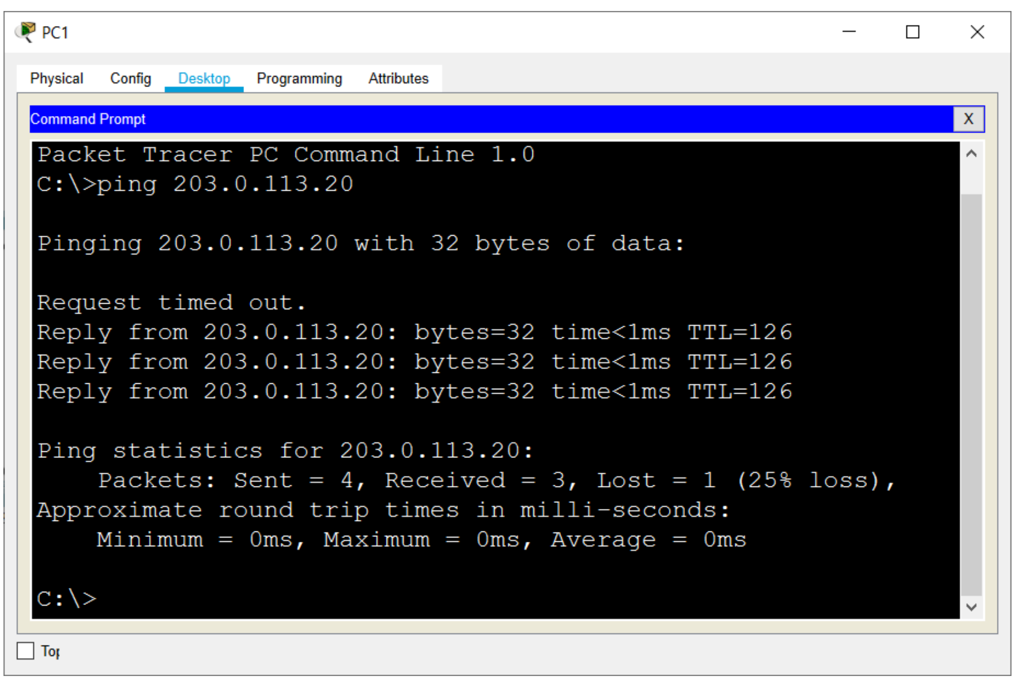


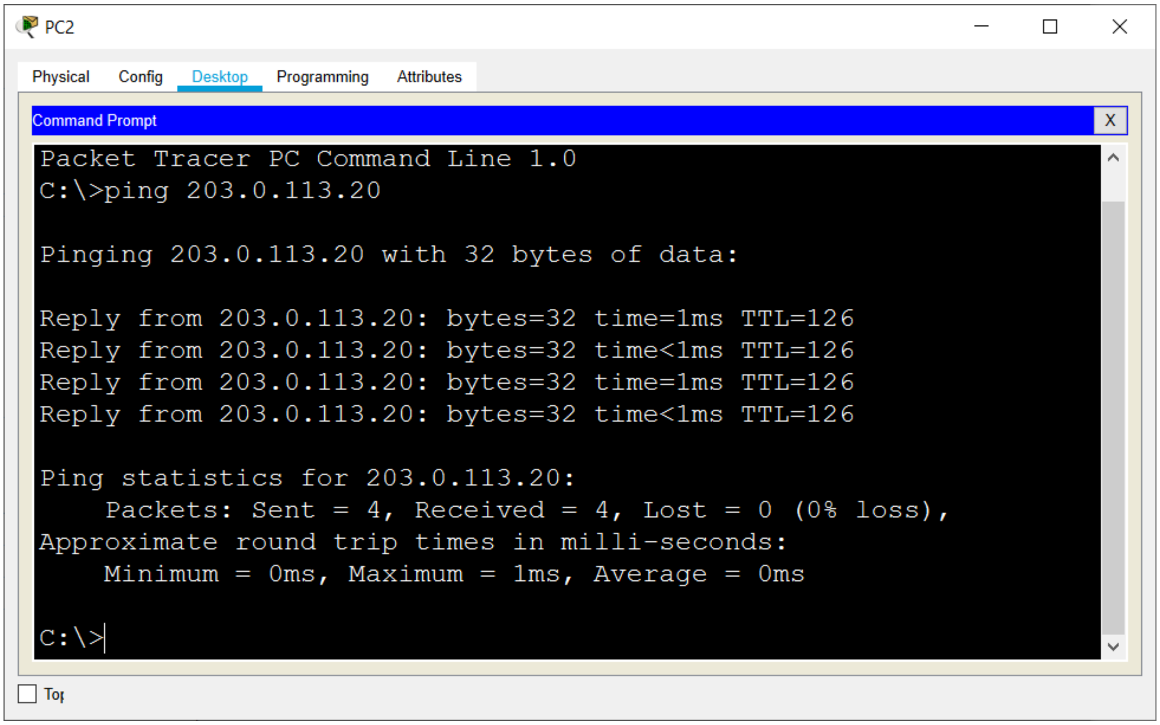


The connection fails.

Next check that PC1 and PC2 can ping EXT-S1.

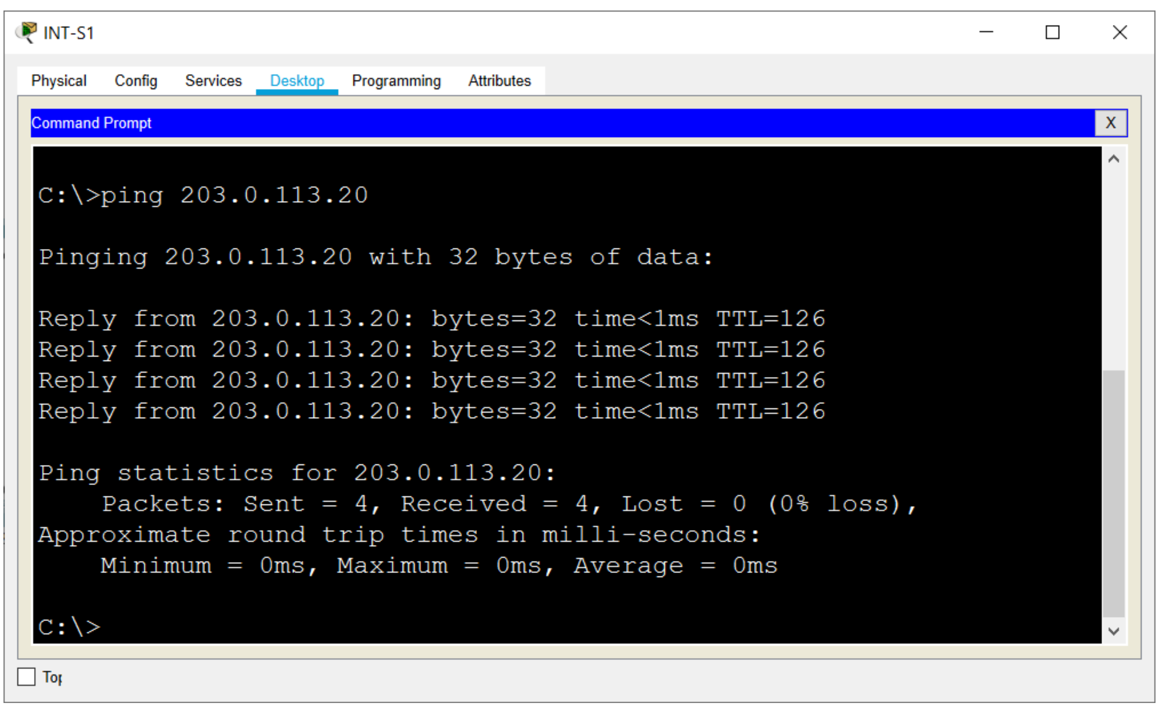






Connectivity from PC1 and PC2 to the Internet is however working.

These symptoms would typically suggest that a static NAT rule for INT-S1 is missing. But a ping from INT-S1 to EXT-S1 works, so something else must be the issue.



Check the NAT configuration on R1 to determine the problem.

```
R1#show run
!
truncated
!
interface FastEthernet0/0
 ip address dhcp
 ip nat outside
!
interface FastEthernet0/1
 ip address 10.0.1.1 255.255.255.0
 ip nat inside
!
interface FastEthernet1/0
 ip address 10.0.2.1 255.255.255.0
 ip nat inside
!
ip nat inside source list 1 interface FastEthernet0/0
overload
ip nat inside source static 10.0.1.10 203.0.113.13
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
access-list 1 permit 10.0.2.0 0.0.0.255
!
truncated
```

The router has not been configured according to the original task. The original task specified:

'203.0.113.2 is to be assigned to the Internet facing F0/0 interface on your router R1.'

The router has however been configured as a DHCP client on its F0/0 interface.

```
interface FastEthernet0/0
 ip address dhcp
```

'A permanent translation should be set up so that external hosts on the Internet can reach the INT-S1 web server using IP address 203.0.113.3.'

A static NAT rule has however been configured to translate the DHCP address on F0/0 (not 203.0.113.13) to the INT-S1 private address 10.0.1.10

```
interface FastEthernet0/0
 ip address dhcp
 ip nat outside
!
interface FastEthernet1/0
 ip nat inside
!
ip nat inside source static 10.0.1.10 203.0.113.13
```

'The 50 internal hosts on the 10.0.2.0/24 internal network should be able to use the remaining public IP addresses to communicate with external hosts. Internal hosts should share public IP addresses where necessary.'

Dynamic NAT with overload has however been configured to translate the DHCP address on F0/0 (not 203.0.113.4 - 12) to the hosts on the 10.0.2.0/24 network.

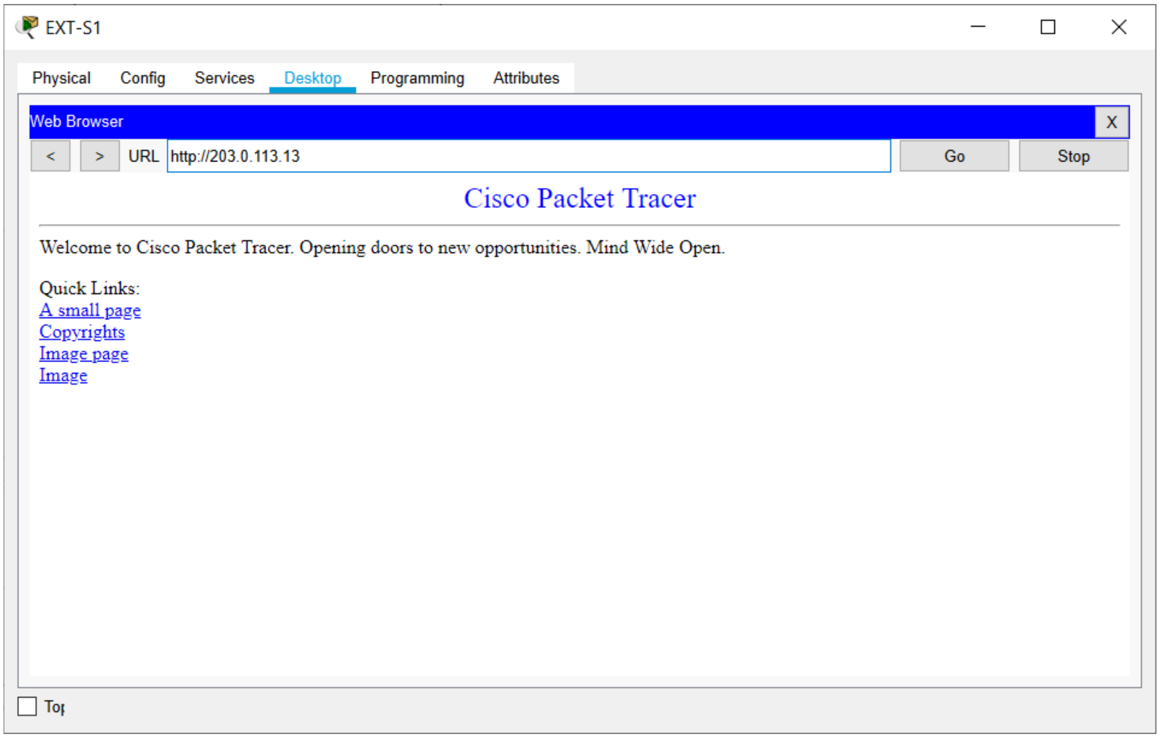
```
interface FastEthernet0/0
 ip address dhcp
 ip nat outside
!
interface FastEthernet2/0
 ip nat inside
!
ip nat inside source list 1 interface FastEthernet0/0
overload
!
access-list 1 permit 10.0.2.0 0.0.0.255
```

Incoming traffic from the Internet to INT-S1 on 203.0.113.3 is not working because R1 has been configured to be a DHCP client on its outside interface rather than using the static IP address 203.0.113.3. Check what IP address R1 has on interface F0/0.

```
R1#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/0        203.0.113.13    YES DHCP   up                    up
FastEthernet0/1        10.0.1.1        YES NVRAM  up                    up
FastEthernet1/0        10.0.2.1        YES NVRAM  up                    up
FastEthernet1/1        unassigned      YES NVRAM  administratively down down
```

(Note that in the real world if a router is expected to have a static IP address then the service provider router will not be configured as a DHCP server. Your router would not be able to get an IP address from DHCP in this case.)

Open the web browser on EXT-S1 again and verify that INT-S1 can receive incoming HTTP connections using the public IP address 203.0.113.13 on R1's F0/0 interface.



Verify NAT translations are in the NAT table on R1 for INT-S1, PC1 and PC2. Translations time out quickly so you may need to send pings from PC1 and PC2 to EXT-S1 again.

```
R1#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.13:1     10.0.2.11:1        203.0.113.20:1     203.0.113.20:1
icmp 203.0.113.13:2     10.0.2.11:2        203.0.113.20:2     203.0.113.20:2
icmp 203.0.113.13:3     10.0.2.11:3        203.0.113.20:3     203.0.113.20:3
icmp 203.0.113.13:4     10.0.2.11:4        203.0.113.20:4     203.0.113.20:4
icmp 203.0.113.13:5     10.0.2.10:5        203.0.113.20:5     203.0.113.20:5
icmp 203.0.113.13:6     10.0.2.10:6        203.0.113.20:6     203.0.113.20:6
icmp 203.0.113.13:7     10.0.2.10:7        203.0.113.20:7     203.0.113.20:7
icmp 203.0.113.13:8     10.0.2.10:8        203.0.113.20:8     203.0.113.20:8
---  203.0.113.13       10.0.1.10          ---                ---
tcp 203.0.113.13:80     10.0.1.10:80       203.0.113.20:1026  203.0.113.20:1026
tcp 203.0.113.13:80     10.0.1.10:80       203.0.113.20:1027  203.0.113.20:1027
```

(NAT translations time out quite quickly. Your output may be different depending on what traffic you have sent through the router recently.)

So NAT is working but it is using the 203.0.113.13 DHCP address on R1's F0/0 interface for all translations, rather than 203.0.113.3 for INT-S1 and 203.0.113.4 – 203.0.113.12 for the hosts on the 10.0.2.0/24 network.)

To correct the issue you need to first remove the erroneous NAT commands. Check the current commands.

```
R1#sh run | section nat
 ip nat outside
 ip nat inside
 ip nat inside
ip nat inside source list 1 interface FastEthernet0/0
overload
ip nat inside source static 10.0.1.10 203.0.113.13
```

Clear the NAT translation table.

```
R1#clear ip nat translation *
```

Remove the static and dynamic NAT rules.

```
R1(config)#no ip nat inside source list 1 interface
FastEthernet0/0 overload
R1(config)#no ip nat inside source static 10.0.1.10
203.0.113.13
```

Replace the DHCP address on R1 F0/0 with the correct static IP address.

```
R1(config)#interface f0/0
R1(config-if)#ip add 203.0.113.2 255.255.255.240
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/0        203.0.113.2     YES manual up                    up
FastEthernet0/1        10.0.1.1        YES NVRAM  up                    up
FastEthernet1/0        10.0.2.1        YES NVRAM  up                    up
FastEthernet1/1        unassigned      YES NVRAM  administratively down down
```

Replacing the IP address will cause the default static route to the SP1 router to be removed. Add it back again.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

The NAT statements on the interfaces, and the access list to specify the internal hosts, have already been configured so do not need to be re-entered.

```
interface FastEthernet0/0
 ip nat outside
!
interface FastEthernet1/0
 ip nat inside
!
interface FastEthernet2/0
 ip nat inside
!
access-list 1 permit 10.0.2.0 0.0.0.255
```

Enter the correct static NAT rule for INT-S1.

```
R1(config)#ip nat inside source static 10.0.1.10
203.0.113.3
```

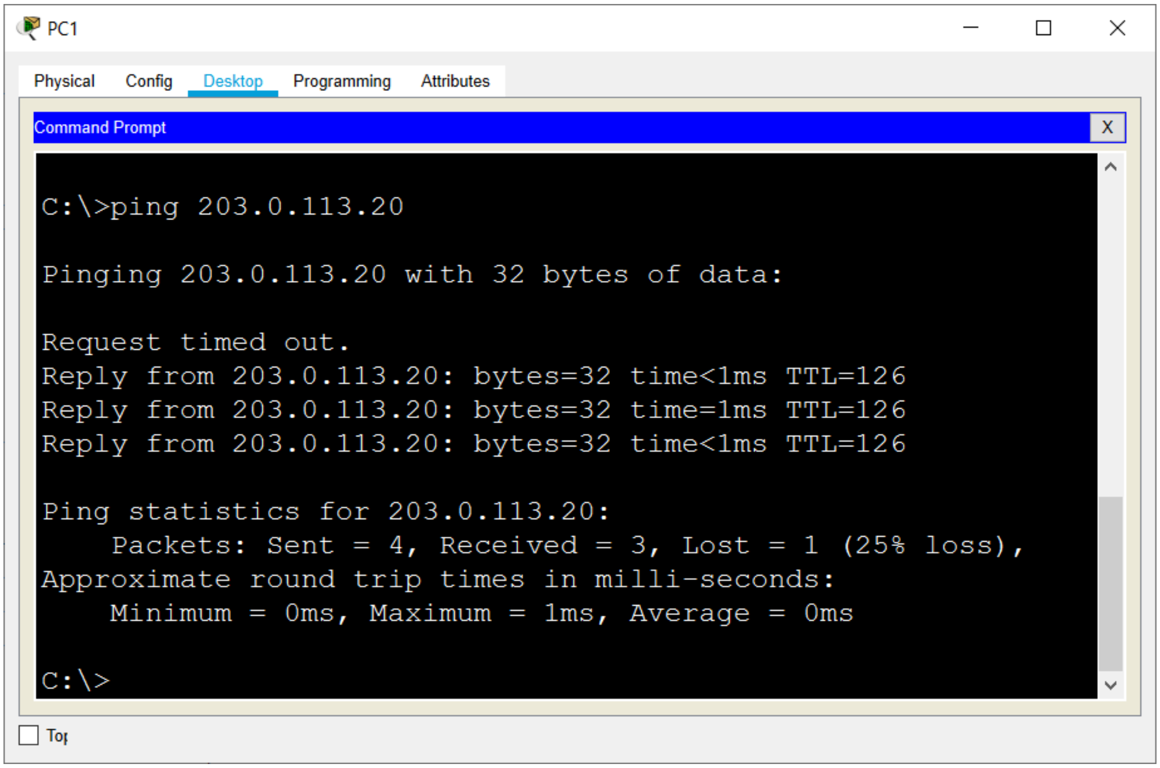
Configure a pool of global addresses for the dynamic NAT rule for the hosts on the 10.0.2.0/24 network.

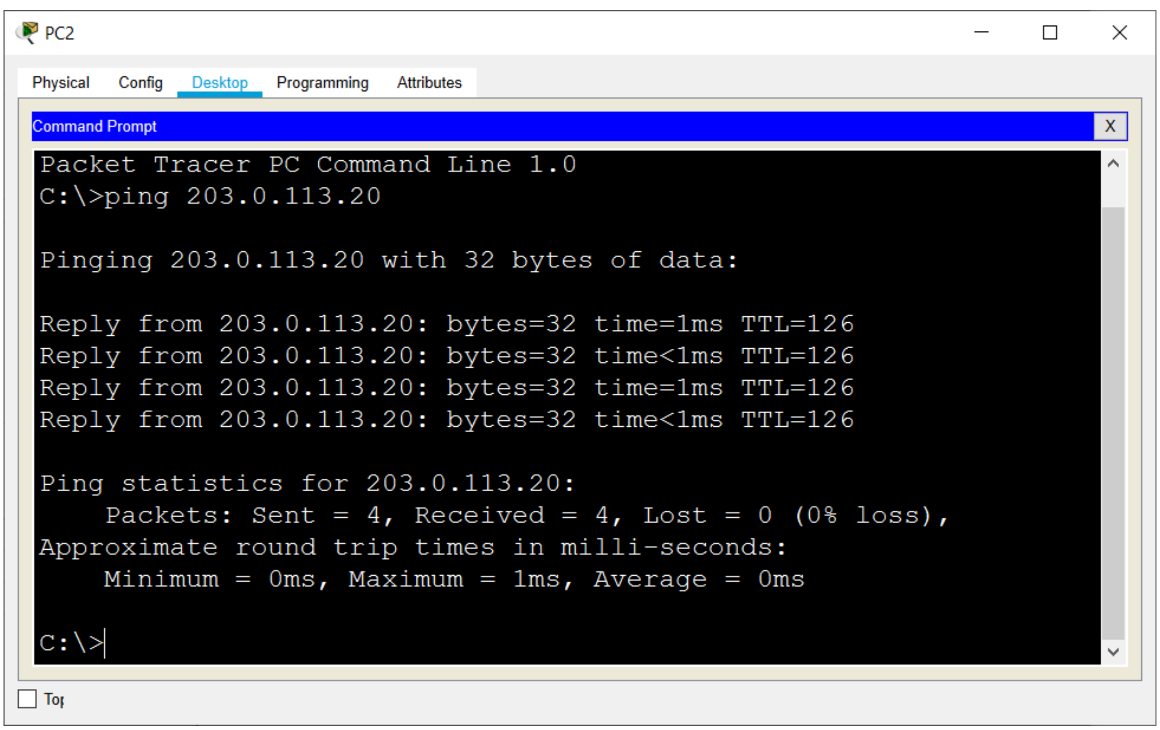
```
R1(config)#ip nat pool Flackbox 203.0.113.4 203.0.113.12
netmask 255.255.255.240
```

Associate the access list with the NAT pool to complete the configuration.

```
R1(config)#ip nat inside source list 1 pool Flackbox
```

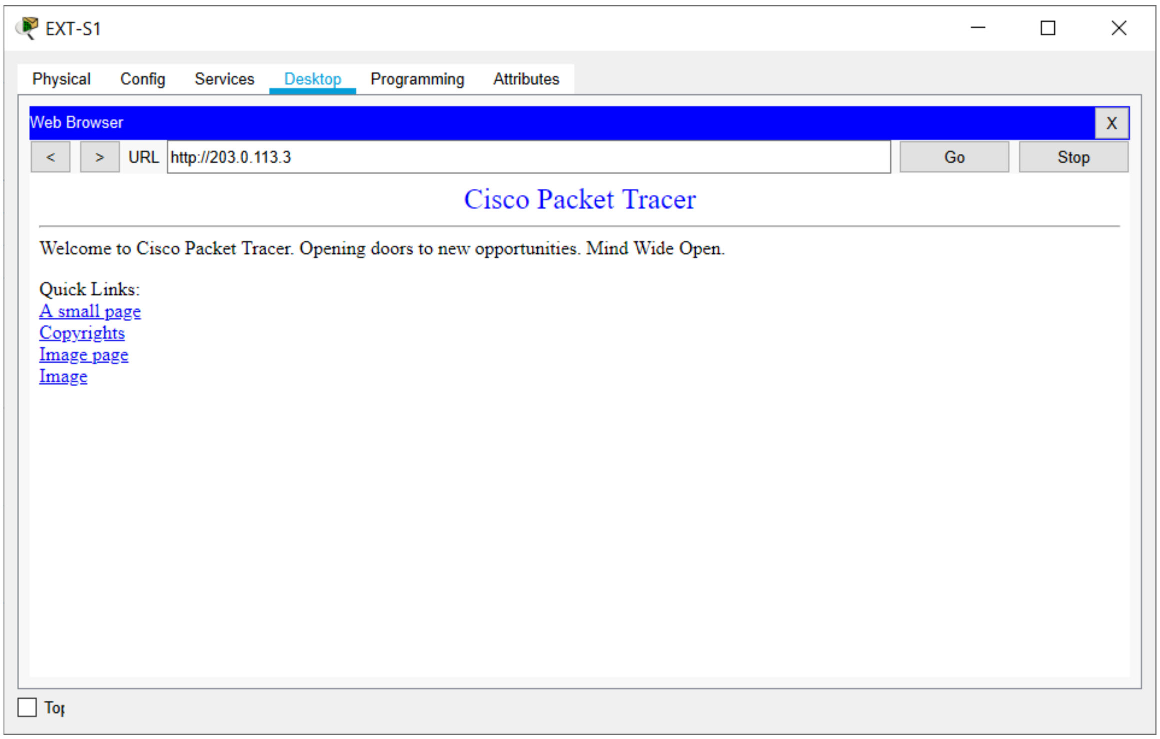
Verify the hosts on the 10.0.2.0/24 network have connectivity to the Internet using the dynamic NAT pool.





```
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.4:5     10.0.2.11:5        203.0.113.20:5     203.0.113.20:5
icmp 203.0.113.4:6     10.0.2.11:6        203.0.113.20:6     203.0.113.20:6
icmp 203.0.113.4:7     10.0.2.11:7        203.0.113.20:7     203.0.113.20:7
icmp 203.0.113.4:8     10.0.2.11:8        203.0.113.20:8     203.0.113.20:8
---  203.0.113.3       10.0.1.10          ---                ---
```

Finally, verify hosts on the Internet can connect to INT-S1 via HTTP to 203.0.113.3




```
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  203.0.113.3       10.0.1.10          ---                ---
tcp 203.0.113.3:80     10.0.1.10:80       203.0.113.20:1029  203.0.113.20:1029
```